

University of Portsmouth

School of Creative Technologies

Final Year Project

How approachable is scientific and mathematical academia to game developers that have little to no formal background

by

Tayler Parsons
UP2212801

Supervisor: Jahangir Uddin

July 29, 2026

Project Type: Study

I consent to my report in this attributed format (not anonymized), subject to final approval by the Board of Examiners, being made available in the Library Dissertation Repository and/or the school digital repositories for a maximum of 10 years.

☒ Yes ☐ No

Modern video game development increasingly relies on complex algorithms, physics simulations, and mathematical concepts. However, it could be argued that a significant gap between contemporary research and the skills of game developers who lack a formal academic background. This report investigates the accessibility of different scientific and mathematical literature from a game development perspective. To evaluate this, this study uses different approaches such as first-hand research that was conducted through presenting developers with standard academic notation such as summation ($\sum$) and product ($\prod$) to assess their baseline comprehension. This report aims to highlight the need for more accessible academic resources to help bridge the gap between the academic and practical side of game development.

Contents

1	Introduction	2
1.1	Defining the developer	2
1.2	Defining the paper	3
2	Literature Review	4
2.1	Introduction	4
2.2	Foundational Mathematics in Game Development	5
2.3	Algorithms & Accessibility	6
2.4	Documentation & Communication	6
2.5	How Do We Determine If a Paper is Accessible?	7
2.5.1	Accessibility Metric System	7
2.5.2	Identified Gaps and What They Imply	8
2.6	Questionnaire	9
2.6.1	Designing the Questionnaire	9
2.6.2	Scoring System	9
3	Methodology	11
3.1	Study Design	11
3.2	Participants	11
3.2.1	Questionnaire	12
3.3	Data Analysis	13
3.3.1	Quantitative Analysis	13
4	Understanding Paper Approachability	19
4.1	Case Studies	19
4.1.1	“A Survey on SPH Methods in Computer Graphics” (Koschier et al., 2022)	19
4.1.2	“Learn OpenGL: Extensive Tutorial Resource for Learning Modern OpenGL” (de Vries, 2014)	20
5	Conclusion	22
5.1	Is there a requirement for a new model of paper?	22
5.2	What would this paper look like?	22

References	24
6 Appendix A: Participant Questionnaire and Responses	25
6.1 Blank Questionnaire	25
6.2 Aggregated Responses and Qualitative Feedback	29

Chapter 1

Introduction

While players primarily experience gameplay and immersive worlds, these experiences are often driven - behind the scenes - by complex algorithms and mathematical concepts.

As the creative and technical vision of the project expand, the developers' reliance on advanced mathematics scales with it; whether it be to achieve more realistic physics, procedural generation, or optimize existing systems. However, when developers turn to contemporary research literature to implement or optimize these visions, they may face a critical roadblock. A significant gap exists between the formal academic notation that can be found in scientific papers, such as summation ($\sum$), or product ($\prod$) symbols, and the practical, programming-orientated skill-set of developers who may lack an existing formal background in these fields of academia. Consequently, the literature that has been created to provide a way to establish and solve problems become arguably inaccessible to a large portion of the development community.

This report aims to measure a wide-range of developers' baseline comprehension of different common-place concepts and notation that can be found in mathematic literature, and comparing it to the level of explanation and context that can be found in these papers.

1.1 Defining the developer

In the context of this report, a 'developer' is someone who:

1. has written code in C++ *or in a comparable game-engine language such as C#*
2. has less than **2 years** of formal further/higher-education in mathematics
3. is currently engaged in either **independent, hobbyist, or industry** game development *or currently engaged in similar fields such as software engineering or web development*

1.2 Defining the paper

In the context of this report, a ‘paper’ may refer to:

1. A report that is written specifically with game development in mind
2. A report that is written with physicists or mathematicians in mind
3. A report from adjacent technical fields, such as computer science, graphics programming or artificial intelligence, which uses any mathematical concepts.
4. Any peer-reviewed academic literature, journal article, or any other formal academic writing that contains contemporary physics or mathematical content.

Chapter 2

Literature Review

2.1 Introduction

Modern game engines such as Unity, Unreal, or custom engines frequently relies on advanced mathematics to achieve various tasks like linear algebra for 3D rendering. However, the academic literature that underlines these methods often presume a specific level of existing understanding in mathematics or physics. This creates a barrier-of-entry for hobbyists, indie developers, or anyone else who lack knowledge in these areas.

The purpose of this literature review is to:

- (i) survey the current state of existing research of mathematical concepts often used in game development
- (ii) evaluate how effectively these papers communicate these concepts to a non-specialist audience
- (iii) identify any gaps that may justify the need for more practical (as in, opposed to practical)-orientated resources

2.2 Foundational Mathematics in Game Development

Table 2.1: Mathematical Concepts Used in Game Development

Concept	Use	Concepts Feature In
Linear Algebra (Vectors, Matrices, Quaternions)	Object transformations, viewport manipulation	“Quaternions, Interpolation and Animation” (Dam et al., n.d., p. 6) <i>3D Math Primer for Graphics and Game Development</i> (Dunn & Parberry, 2011)
Differential Calculus (Derivatives, Gradients)	Physics engines, procedural animation	“Real-Time Inverse Kinematics for Generating Multi-Constrained Movements of Virtual Human Characters” (Voss & Kopp, 2025, p. 3) “A Survey on SPH Methods in Computer Graphics” (Koschier et al., 2022)
Probability and Statistics	AI decision-making, procedural generation	“Learning to Generate Video Game Maps Using Markov Models” (Snodgrass & Ontañón, 2017) “Learning Probabilistic Behavior Models in Real-Time Strategy Games” (Dereszynski et al., 2011)
Vector Calculus	Particle simulations	“A Survey on SPH Methods in Computer Graphics” (Koschier et al., 2022) “Fluid Simulation with the Navier-Stokes Equations” (Wang, n.d.)

Whilst being the most popular of these concepts, Linear Algebra’s most ‘basic’ operations such as matrix multiplications are often represented with $\text{\LaTeX}$ notation that developers must manually translate into code. Few papers, such as *Unity Artificial Intelligence Programming* (Aversa, 2022) attempt to bridge this gap by providing code examples, however the paper still assumes a level of fluency with the mathematical concepts that many indie developers may lack.

2.3 Algorithms & Accessibility

Table 2.2: Algorithms and Accessibility in Papers

Technique	Representation	Alternatives
Procedural Generation	Noise functions are expressed through the use of summation notation (e.g., $\sum P(i)$)	Unity’s Perlin Noise API. Documented resources such as tutorials (e.g., “Unity3D Procedural Generated Low-Poly Terrain” (Glazycheva, 2020))
Physics Engine	Equations of motion expressed through the use of differential notation ($\partial^2 x / \partial t^2 = F/m$)	Bullet, PhysX, Jolt. Most documentation omit the fundamental mathematics. “Jolt Physics: Jolt Physics” (“Jolt Physics: Jolt Physics”, n.d.)
AI Pathfinding	Expressed through Markov Chain ($\pi(\theta)$) and matrix notation	A* and Dijkstra’s algorithms are very popular, and often omit the probability theory that make them operate

Whilst academic papers present the mathematics in its most unadulterated form, communities have developed standalone libraries that abstract most of the complexity. However, these libraries are rarely accompanied by the theoretical context beside it, which would help a developer understand *how* an algorithm works, which may limit their ability to expand or optimize the code.

2.4 Documentation & Communication

Table 2.3: Documentation and Communication Practices in Papers

Source	Strengths	Weaknesses
Peer-Reviewed Journals	Very Comprehensive	Very dense in notation, little to no code examples
Conferences	Primarily focuses on practical implementations, live demos	Lacks peer review, may over simplify the mathematics in sake of brevity
Industry Blogs & Tutorials	Code-focused, step-by-step explanations	Usually limited in theoretical explanation, and can be outdated
Open-Source Projects	Very accessible code, community contribution / feedback	Documentation can be inconsistent or lacking

These formats often excel at their strengths, but lack the structure to convey that which they lack. Due to this, it is quite hard to find a source that achieves to provide a structured

synthesis between the mathematical notation and code. This disconnect is what prevents the vast majority of the sources in the categories to be accessible.

2.5 How Do We Determine If a Paper is Accessible?

2.5.1 Accessibility Metric System

Definitions of Concept Approachability Dimensions

We can determine approachability for any given paper through categorising each ‘*dimension*’ of approachability:

Table 2.4: Concept Approachability

Dimension	Definition	Indicator Example
Notation	To which degree are mathematical symbols explained or avoided	A ‘notation glossary’ maybe present
Clarity of Lexicon	To which degree is plain language use, and whether technical jargon is avoided	Number of technical terms in any given chapter / section of the paper
Contextualisation	To which degree do papers provide examples, diagrams, or code snippets to help provide context to the topic discussed	At least one code snippet
Pedagogy	To which degree do papers aim to teach the reader about a specific topic	Any attempt to help teach the reader what that concept or notation is

Note. The difference between contextualisation and pedagogy lies in intent: contextualisation provides clarity to the reader on *what* a concept is (e.g., through examples or diagrams), where as pedagogy involves an active attempt to *teach* the reader how and why it works.

Each dimension will be scored on a 0-4 scale (0 = no accessibility, 4 = very accessible). The overall *Approachability Score* (AS) will be the mean value of the four sub-scores for that paper.

Table 2.5: Literature Approachability

Source	Contribution to Criteria
<i>Unity Artificial Intelligence Programming</i> (Aversa, 2022)	Features code comments, but expects prior knowledge of the basics of AI theory
“Quaternions, Interpolation and Animation” (Dam et al., n.d.)	Frequently uses advanced mathematical notation with very little explanatory text
“A Survey on SPH Methods in Computer Graphics” (Koschier et al., 2022)	Very dense advanced mathematical notation with few diagrams or explanatory text
“Real-Time Inverse Kinematics for Generating Multi-Constrained Movements of Virtual Human Characters” (Voss & Kopp, 2025)	Whilst still dense in mathematical notation, it provides some indicators of what some values and parts of equations mean

These studies suggest that **notation** and **contextualisation** are the most frequently neglected dimensions. However, in contrast, the *lexical clarity* tend to be adequate in industry blogs but is rarely ever present in peer-review papers.

2.5.2 Identified Gaps and What They Imply

1. **Notation Barrier:**
2. **Lack of Code Examples:**
3. **Lack of Pedagogy:**

2.6 Questionnaire

2.6.1 Designing the Questionnaire

The questionnaire will be informed by the gaps identified above. It will feature questions such as:

Table 2.6: Questionnaire Example Questions

Question	Rationale
How comfortable are you translating mathematical equations (e.g., $\partial^2 x / \partial t^2$) into code?	Evaluates the ‘lack of code examples’ gap
When reading academic papers, how often do you need to look up definitions for the symbols and concepts featured?	Evaluates the notation dimension described in table 2.4
Does the presence of pseudocode or code snippets increase your confidence in understanding a given algorithm or concept?	Validates the contextualisation dimension described in table 2.4
Do you find that visual diagrams (such as those that show coordinate spaces, or vector trajectories) are more informative than formal mathematical definitions when learning a new algorithm	Investigates the pedagogy dimension described in table 2.4
When using a physics or path-finding library, (e.g., Jolt, A*), how important is it to understand the underlying mathematical theory vs. just using the API?	Addresses the trade-offs mentioned in Section 3 regarding abstraction and theoretical contexts

Each item will use a 5-point Likert scale (1 = Not at all comfortable, 5 = Very Comfortable). The questionnaire will also collect some demographic data (such as years of coding experience, formal maths training) to determine any background variables that may influence the results.

2.6.2 Scoring System

For each paper examined in the literature review, each paper will be evaluated to determine how it approaches the four dimensions (Notation, Clarity of Lexicon, Contextualisation, Pedagogy) using the 0-4 scale. The mean of these scores will constitute the **Approachability Score (AS)**:

$$AS = \frac{1}{4} \sum_{d=1}^4 S_d \quad (2.1)$$

Where S_d is the sub-score for dimension d . The questionnaire data will then be cross-referenced with the AS. The papers with low scores should correspond to lower comprehension

of the developers. This would validate the scoring system.

Chapter 3

Methodology

3.1 Study Design

To investigate how approachable scientific and mathematical literature is for game developers, a study was conducted. This study, primarily a questionnaire provides quantitative statistics into the level of comprehension and the attitudes that game developers may possess when reading, or not reading, scientific and mathematic literature, while open-ended questions provide a qualitative perspective where uses are allowed to provide personal anecdotes of their experiences with scientific and mathematical academia.

3.2 Participants

Posts were sent out on platforms such as Discord and LinkedIn, requesting that game developers, primarily programmers fill out the questionnaire. Due to time scheduling concerns, the survey only lasted around one week. Due to this, the participant pool was quite low; 15 unique respondents.

All the respondents met all the inclusion criteria that was established in the posts, and in the questionnaire itself.

3.2.1 Questionnaire

The questionnaire (*see Appendix A*) comprises:

Table 3.1: Questionnaire

Section	Items	Purpose
Demographics	1-3	Determine background variables that indicate comprehension (e.g., the highest level of formal mathematics qualification)
Paper Readers	4-7	Evaluate how participant interact, and respond to, notation, rate the clarity of lexicon on different mediums (papers vs. blogs)
Barriers to Entry	8-9	Identify reasons for avoiding academic literature
Code vs. Notation	12-12	Determine confidence in translating standard notation ($\sum$) into code

3.3 Data Analysis

3.3.1 Quantitative Analysis

Demographics

Table 3.2: Demographics Results

Characteristic	Category	Count	Percentage
Primary Background	Student	9	60%
	Working in Industry	3	20%
	Indie	2	13%
	Hobbyist	1	7%
	In an adjacent field	0	0%
Maths Education Level	GCSE	7	47%
	Higher Education	6	40%
	A-Levels	2	13%
	Self-Taught	0	0%
	Other	0	0%
How often do you refer to formal or informal academic media?	Frequently	10	67%
	Occasionally	4	27%
	Rarely	1	7%
	Never	0	0%

Various points can be extrapolated from the data listed in table 3.2, such as:

- The majority of respondents identified themselves as currently being students

- A substantial majority (**67%**) report referring to formal or informal academic media *frequently*
- The largest singular reported educational level is GCSE (**47%**). Although further education *A-Levels and Higher Education* also posses a combined significant presence, the dominance of GCSE-educated respondents might suggest that within a large sample pool, the majority might not have completed advanced degrees.
- Due to the point above, it remains evident that academic content needs to be highly accessible.

Paper Readers

Table 3.3: Paper Readers Results

Question	Answer	Count	Percentage
When reading a paper, how do you interact with mathematical notation (e.g., $\sum$, $\prod$)?			
	I look up their definitions on external sources	6	43%
	I understand them comfortably	4	29%
	I skip past the maths and try look for code snippets	2	14%
	I rely on using AI to explain and give context to the notation used	1	7%
	I do not understand some of the notation used	1	7%
How would you rate the average clarity of the language used in peer-reviewed papers vs. blogs or other sources?			
<i>a) Peer-Reviewed Papers</i>			
	Confusing	8	57%
	Dense	3	21%
	Clear	2	14%
	Understandable	1	7%
<i>b) Industry blogs / other sources</i>			
	Clear	9	64%
	Understandable	4	27%
	Dense	1	7%
	Confusing	0	0%
Which of the following elements add the most value when attempting to implement a complex algorithm from a paper?			

Table 3.3: Paper Readers Results

Question	Answer	Count	Percentage
	All of the above	7	24%
	High level pseudo-code	6	21%
	Fully implemented code snippets	6	21%
	Step-by-step diagrams	6	21%
	Formal writing	4	14%
	None of the above	0	0%
	Other	0	0%
How often do you find that a paper attempts to TEACH you what a concept is, instead of just providing clarification?			
	Rarely	8	57%
	Occasionally	5	36%
	Never	1	7%
	Frequently	0	0%

Based the section of the survey detailed in table 3.3, various key points can be extrapolated. Such as:

- **Peer-reviewed papers have clarity issues**

- 78% of respondents described peer-reviewed papers as **confusing** or **dense**; only **21%** described them as **clear** or **understandable**
- In contrast, **91%** described industry blogs and other secondary sources as **Clear**, or **Understandable**, with zero respondents finding them confusing Given these points, it is reasonable to suggest that readers find informal sources more approachable than formal academic papers

- **Notation is a large barrier**

- Only **29%** of respondents say that they could comfortably read mathematical notation directly within a paper
- **71%** said that they required external aids, or information, such as:
 - * **43%** look up definitions for notation and syntax on external sources
 - * **14%** immediately skip all mathematics and text to find code snippets
 - * A further **14%** said that they rely on AI tools to explain the notation, our that the outright do not understand it

Given these points, it is reasonable to assume that dense mathematical notation acts as a bottleneck for most readers, requiring them to leave the paper to seek context from external sources

- **Readers often prefer artefacts to writing**

When it comes to implementing algorithms or mathematical equations, readers prefer practical tools than formal writing:

Barriers to Entry

Table 3.4: Barriers to Entry Results

Question	Answer	Count	Percentage
What are the primary reasons that you do not use academic literature for game development?			
	I feel like I do not need to use academic papers	1	20%
	The papers and their contents are daunting (notation, context, etc.)	1	20%
	Too much academic jargon	1	20%
	I feel like there is a ‘disconnect’ from the theory in the paper and the implementation in engine	1	20%
	I prefer tutorials, blogs, or other sources	1	20%
	I do not know how to find relevant papers	0	0%
If academic literature consistently used functional code samples alongside their mathematical content, how likely would you be to use them?			
	Neither likely nor unlikely	1	100%
	Likely	0	0%
	Likely	0	0%

Despite only one respondent answering this section of the survey detailed in table 3.4, various key points can be extrapolated, such as:

- **Avoidance of papers is a multi-faceted problem rather than an individual issue** The respondent selected 5 out of the 6 listed reasons for avoiding academic literature. This might suggest that for other developers that abstain, might do so due to multiple reasons:
 - **Lack of Relevance:** Some may perceive that academic literature simply might not be relevant for their work or use case
 - **Comprehension:** Jargon and mathematical notation maybe considered daunting and inaccessible
 - **Disconnect between theoretical and practical:** There may be a disconnect and difficulty translating the theoretical concepts detailed in the paper into a practical environment
 - **Preference for alternative sources:** Alternative sources may be preferred such as tutorials, blogs, or other less formal sources

- **Discovery and Access may not be a major issue**

The respondent said that they do not struggle with finding or accessing papers, however this may be due to their lack of interest in doing so.

Code vs. Notation

Table 3.5: Respondents Equation Solving Confidence

Participant	1st Choice	2nd Choice	3rd Choice
1	No idea where to start	Confident	Somewhat confident
2	Confident	No idea where to start	Somewhat confident
3	No idea where to start	Somewhat confident	Confident
4	Somewhat confident	No idea where to start	Confident
5	No idea where to start	Somewhat confident	Confident
6	Confident	Somewhat confident	No idea where to start
7	Somewhat confident	No idea where to start	Confident
8	Confident	Somewhat confident	No idea where to start
9	Somewhat confident	No idea where to start	Confident

Table 3.6: Summary of Response Choices

Confidence Level	1st Choice		2nd Choice		3rd Choice	
	<i>count</i>	%	<i>count</i>	%	<i>count</i>	%
No idea where to start	3	33	4	44	2	22
Somewhat confident	3	33	2	22	4	44
Confident	3	33	3	33	3	33
Total (<i>count</i>)	9		9		9	

Table 3.7: Underlying Concepts Importance

Question	Answer	Count	Percentage
When using native game engine features such as Unity's Perlin noise functions, how important is it to you that you understand the underlying concepts and mathematics behind them?	Somewhat important	5	33%
	Neutral	4	27%
	Extremely important	3	20%
	Somewhat unimportant	3	20%

Table 3.7: Underlying Concepts Importance (Continued)

Question	Answer	Count	Percentage
	Not important	0	0%

Based on the data shown across tables 3.5, 3.6, and 3.7, several observations can be made:

- **Polarized Problem-Solving Confidence:**

- Each level of confidence was selected as the first choice by exactly 3 participants each **33.3%**
- Over two thirds of the participants ($6/9$) landed at the extreme ends of the spectrum for their first choice. Either feeling completely helpless, or fully confident

- **Respondents Broadly Prefer to Implement Concepts, Rather than Understanding Them:**

When asking developers how important it is to understand the mathematics and concepts behind built-in engine tools such as Unity’s Perlin Noise:

- Respondents indicated that it was neutrally important, to extremely important to understand them, none stated that it was completely unimportant
- The disparity between the groups of people that view understanding the concepts as ‘somewhat important’ to neutral, suggests that developers desire *just* enough context to allow them to understand, modify, or make use of the concepts rather than fully understanding the mathematical proofs behind them

Chapter 4

Understanding Paper Approachability

Findings across multiple questionnaire sections suggest that there is a fundamental disconnect in scientific literature with game development. While the survey participants showed high levels of curiosity and intrigue, this interest is also being constrained by the daunting nature of the academia.

The data suggests a contradiction. Developers are constantly looking for knowledge to do what they need, yet they lack the immediate tools and requisites to process it. This is evidenced through the high reliance on external sources (**43%** look up definitions) alongside the noticeable percentage that skips past the mathematics to look for code snippets (**14%**). The current academic environment requires the developer to become an amateur mathematician or physicist in order to start programming.

The contrast between professional journals (**57% Confusing/Dense**) and blogs (**91% Clear/Understandable**) indicates two different educational ecosystems. Where academic papers are built primarily for specialist-to-specialist communication, whilst developers seem to prefer a less rigorous communicative approach. This suggests that formal literature fails at its primary purpose of clear dissemination of information.

The reluctance to translate notation in to code highlights a core barrier. Theoretical concepts are presented in an abstract, formal manner that does not map well into practical logic. The mathematics is taught as a concept (*what*) rather than as a process (*how*)

4.1 Case Studies

To illustrate the inherent challenges in current academic literature, a selection of papers were analysed.

4.1.1 “A Survey on SPH Methods in Computer Graphics” (Koschier et al., 2022)

The structure of this paper arguably severely limits the comprehension when looking at it from an approachability view.

When scoring the paper against our four **Accessibility Dimensions** it scored:

1. **Notation:** The text heavily relies on specialised abstract mathematical notation which are never properly defined for a non-specialist, scoring **0/4**
2. **Clarity of Lexicon:** The jargon used (e.g., ‘Smoothed Particle Hydrodynamics’, and ‘Lagrangian frame’) assumes that the reader has an existing advanced knowledge of the content, scoring **1/4**
3. **Contextualisation:** While diagrams exist, they are mathematically complex and do not provide functional code examples, instead offering highly-abstracted pseudocode, scoring **2/4**
4. **Pedagogy:** The paper functions as a survey *for specialists* rather than a teaching tool. It assumes existing foundational knowledge meaning that it does not intend to teach the reader, scoring **0/4**

The resulting **Approachability Score** (AS) for this type of literature is therefore:

$$AS = \frac{0 + 1 + 2 + 0}{4} = 0.75/4 \quad (4.1)$$

This very low score reinforces the preliminary findings that, while valuable to researchers, the paper functions as a profound barrier for the average game developer

4.1.2 “Learn OpenGL: Extensive Tutorial Resource for Learning Modern OpenGL” (de Vries, 2014)

When scoring this blog against the **Accessibility Dimensions** it scored:

1. **Notation:** Concepts like matrices, vectors, etc., are never presented purely abstractly. They are shown being **used** within a function signature or variable declaration in code (e.g., `mat3 modelMatrix`), it scores **3/4**
2. **Clarity of Lexicon:** The language, whilst technical, is often framed in functional terms. Instead of purely abstracted mathematical jargon, it uses terminology such as ‘View Matrix’, ‘Projection Pipeline’, it scores **3/4**
3. **Contextualisation:** Every concept has:
 - (a) An explanation of the theoretics (*what it is*)
 - (b) Diagrams showing data or other information
 - (c) Fully functional, commented code snippets that implement the theory discussedit scores **4/4**
4. **Pedagogy:** Due to it being *focused* on being a learning resource, it excels in attempting to *teach* the reader the concepts discussed. It structures the teaching resources so that the reader incrementally learns new information, whilst still being able to understand the context of the text. It scores **3/4**

The resulting **Approachability Score** (AS) for this type of literature is therefore:

$$AS = \frac{3 + 3 + 4 + 3}{4} = 3.25/4 \quad (4.2)$$

This is a very high score, and is a great learning resource for new and returning developers. It remains evident why it is considered the go-to resource for OpenGL graphics programming.

Chapter 5

Conclusion

Based on the evidence, discussed in previous chapters, it is reasonable to presume that given the high engagements of developers, their low comprehension confidence, and a clear preference for *accessible* sources, the conclusion can be made that no single existing academic medium, whether formal or informal, can effectively serve the needs of the game development community.

5.1 Is there a requirement for a new model of paper?

Whilst the current academic ecosystem seems to be functioning as is, the argument could, and perhaps *should* be made whether a new 'model' of paper standard should be proposed. This new model could provide a hybrid approach where the rigour, structure, and formality of peer-reviewed papers is married to the clarity and practical approach of blogs, or other less formal types of academia.

5.2 What would this paper look like?

This new model's function should aim to actively *teach* its concepts using different learning techniques, and should perhaps contain:

- **Theoretical Context:** A concise summary of the core mathematics
- **Visualisation:** Step-by-step (*or at the least, more clear and informative*) diagrams that help map abstract concepts into more tangible, real-world examples
- **Implementations:** Functional, and annotated pseudocode that help translate the theory and concepts discussed

The necessity of a new model can be argued. It should be noted that it is unreasonable to expect all academia, to adopt a new standard of paper, however papers and specific fields that have a large game-developer audience may benefit.

List of Tables

2.1	Mathematical Concepts Used in Game Development	5
2.2	Algorithms and Accessibility in Papers	6
2.3	Documentation and Communication Practices in Papers	6
2.4	Concept Approachability	7
2.5	Literature Approachability	8
2.6	Questionnaire Example Questions	9
3.1	Questionnaire	12
3.2	Demographics Results	13
3.3	Paper Readers Results	14
3.3	Paper Readers Results	15
3.4	Barriers to Entry Results	16
3.5	Respondents Equation Solving Confidence	17
3.6	Summary of Response Choices	17
3.7	Underlying Concepts Importance	17

References

- Aversa, D. D. (2022, March 28). *Unity Artificial Intelligence Programming: Add powerful, believable, and fun AI entities in your game with the power of Unity*. Packt Publishing Ltd.
- Dam, E. B., Koch, M., & Lillholm, M. (n.d.). Quaternions, Interpolation and Animation.
- de Vries, J. (2014). *Learn OpenGL: Extensive tutorial resource for learning Modern OpenGL*. LearnOpenGL. Retrieved July 29, 2026, from <https://learnopengl.com/>
- Derezynski, E., Hostetler, J., Fern, A., Dietterich, T., Hoang, T.-T., & Udarbe, M. (2011). Learning Probabilistic Behavior Models in Real-Time Strategy Games. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, 7(1), 20–25. <https://doi.org/10.1609/aiide.v7i1.12433>
- Dunn, F., & Parberry, I. (2011, November 2). *3D Math Primer for Graphics and Game Development*. CRC Press.
- Glazycheva, E. (2020, August 6). *Unity3D procedural generated low-poly terrain*. Medium. Retrieved July 1, 2026, from <https://medium.com/@glazychevaeo/unity3d-procedural-generated-low-poly-terrain-d11914ab9f71>
- Jolt Physics: Jolt Physics*. (n.d.). Retrieved July 1, 2026, from <https://jrouwe.github.io/JoltPhysics/>
- Koschier, D., Bender, J., Solenthaler, B., & Teschner, M. (2022). A Survey on SPH Methods in Computer Graphics. *Computer Graphics Forum*, 41(2), 737–760. <https://doi.org/10.1111/cgf.14508>
- Snodgrass, S., & Ontañón, S. (2017). Learning to Generate Video Game Maps Using Markov Models. *IEEE Transactions on Computational Intelligence and AI in Games*, 9(4), 410–422. <https://doi.org/10.1109/TCIAIG.2016.2623560>
- Voss, H., & Kopp, S. (2025). Real-Time Inverse Kinematics for Generating Multi-Constrained Movements of Virtual Human Characters. *Proceedings of the 25th ACM International Conference on Intelligent Virtual Agents*, 1–10. <https://doi.org/10.1145/3717511.3747066>
- Wang, A. (n.d.). Fluid Simulation with the Navier-Stokes Equations.

Chapter 6

Appendix A: Participant Questionnaire and Responses

This appendix presents the questionnaire that was created to evaluate game developer demographics and the approachability of academic notation, followed by the data generated from the aggregated responses and qualitative feedback

6.1 Blank Questionnaire

Demographics & Backgrounds

1. What is your primary background in game development?

- Indie
- Hobbyist
- Working in Industry
- Student
- I am in an adjacent field (Software Engineering, IT, etc)

2. What is your highest level of formal mathematics education?

- GCSE
- A-Levels
- Higher Education (Under/Postgraduate)
- Self Taught
- (Other)

3. How often do you refer to formal or informal academic media? (Research Papers, Conferences, Blog Posts, YouTube Videos etc.)

- Never
- Rarely
- Occasionally
- Frequently

Paper Readers

4. [Notation] When reading a paper, how do you interact with with mathematical notation such as (e.g., $\sum$, $\prod$)

- I do not understand some of the notation used
- I rely on using AI to explain and give context to the notation used
- I skip past the maths and try look for code snippets
- I look up their definitions on external sources (e.g other papers, google, etc)
- I understand them comfortably

5. [Lexicon] How would you rate the average clarity of the language used in peer-reviewed papers vs. blogs or other sources

Item	Dense	Confusing	Clear	Understandable
Peer-Reviewed Papers	☐	☐	☐	☐
Industry blogs / other sources	☐	☐	☐	☐

6. [Contextualisation] Which of the following elements add the most value when attempting to implement a complex algorithm from a paper (Select all that Apply)

- High level psuedo-code
- Fully implemented code snippets (C++, C#, etc.)
- Step-by-step diagrams
- Formal writing
- All of the above
- None of the above

- (Other)

7. [Pedagogy] How often do you find that a paper attempts to TEACH you what a concept is, instead of just providing clarification

- Never
- Rarely
- Occasionally
- Frequently

Barriers to Entry

You answered that you rarely, if ever refer to academic literature, why?

8. What are the primary reasons that you do not use academic literature for game development? (Select all that Apply)

- I feel like I do not need to use academic papers
- I do not know how to find relevant papers
- The papers and their contents are daunting (notation, context, etc.)
- Too much academic jargon
- I feel like there is a ‘disconnect’ from the theory in the paper and the implementation in engine
- I prefer tutorials, blogs, or other sources

9. If academic literature consistently used functional code samples alongside their mathematical content, how likely would you be to use them?

- Unlikely
- Neither likely or unlikely
- Likely

Code vs. Notation

10. Look at the following expression. It can be frequently found in literature about procedural generation. How confident are you that you could accurately rewrite this into a for loop in your preferred programming language WITHOUT looking it up?

$$P(x) = \sum_{i=1}^n f(i)$$

- No idea where to start
- Somewhat confident
- Confident

11. When using native game engine features such as Unity’s Perlin noise functions, how important is it to you that you understand the underlying concepts and mathematics behind them?

- Not important
- Somewhat unimportant
- Neutral
- Somewhat important
- Extremely important

12. Any further thoughts?

(Text field)

6.2 Aggregated Responses and Qualitative Feedback

This section presents the raw data counts and the qualitative feedback collected through the questionnaire detailed in Appendix A.

Demographics & Backgrounds

1. What is your primary background in game development?

- Indie (2 responses)
- Hobbyist (1 response)
- Working in Industry (3 responses)
- Student (9 responses)
- I am in an adjacent field (Software Engineering, IT, etc) (0 responses)

2. What is your highest level of formal mathematics education?

- GCSE (8 responses)
- A-Levels (2 responses)
- Higher Education (Under/Postgraduate) (5 responses)
- Self Taught (0 responses)
- (Other) (0 responses)

3. How often do you refer to formal or informal academic media? (Research Papers, Conferences, Blog Posts, YouTube Videos etc.)

- Never (0 responses)
- Rarely (1 response)
- Occasionally (4 responses)
- Frequently (10 responses)

Paper Readers

4. [Notation] When reading a paper, how do you interact with with mathematical notation such as (e.g., $\sum$, $\prod$)

- I do not understand some of the notation used (1 response)
- I rely on using AI to explain and give context to the notation used (1 response)

- I skip past the maths and try look for code snippets (2 responses)
- I look up their definitions on external sources (e.g other papers, google, etc) (7 responses)
- I understand them comfortably (4 responses)

5. [Lexicon] How would you rate the average clarity of the language used in peer-reviewed papers vs. blogs or other sources

Item	Dense	Confusing	Clear	Understandable
Peer-Reviewed Papers	4	8	1	1
Industry blogs / other sources	1	0	9	4

6. [Contextualisation] Which of the following elements add the most value when attempting to implement a complex algorithm from a paper (Select all that Apply)

- High level psuedo-code (5 responses)
- Fully implemented code snippets (C++, C#, etc.) (5 responses)
- Step-by-step diagrams (4 responses)
- Formal writing (3 responses)
- All of the above (6 responses)
- None of the above (0 responses)

7. [Pedagogy] How often do you find that a paper attempts to TEACH you what a concept is, instead of just providing clarification

- Never (1 response)
- Rarely (9 responses)
- Occasionally (4 responses)
- Frequently (0 responses)

Barriers to Entry

You answered that you rarely, if ever refer to academic literature, why?

8. What are the primary reasons that you do not use academic literature for game development? (Select all that Apply)

- I feel like I do not need to use academic papers (1 response)

- I do not know how to find relevant papers (0 responses)
- The papers and their contents are daunting (notation, context, etc.) (1 response)
- Too much academic jargon (1 response)
- I feel like there is a ‘disconnect’ from the theory in the paper and the implementation in engine (1 response)
- I prefer tutorials, blogs, or other sources (1 response)

9. If academic literature consistently used functional code samples alongside their mathematical content, how likely would you be to use them?

- Unlikely (0 responses)
- Neither likely or unlikely (1 response)
- Likely (0 responses)

Code vs. Notation

10. Look at the following expression. It can be frequently found in literature about procedural generation. How confident are you that you could accurately rewrite this into a for loop in your preferred programming language WITHOUT looking it up?

$$P(x) = \sum_{i=1}^n f(i)$$

- No idea where to start (8 responses)
- Somewhat confident (3 responses)
- Confident (4 responses)

11. When using native game engine features such as Unity’s Perlin noise functions, how important is it to you that you understand the underlying concepts and mathematics behind them?

- Not important (0 responses)
- Somewhat unimportant (3 responses)
- Neutral (6 responses)
- Somewhat important (4 responses)
- Extremely important (2 responses)

12. Any further thoughts?

Note: Due to these being direct quotations taken from the questionnaire, these answers are presented verbatim. They have not been modified for spelling, grammar, or other reasons

- *Respondent 6:* “i find myself wanting to understand underlying concepts mostly for optimisation reasons, if i understand the concepts i can sometimes use a cheaper approximation to a given algorithm”
- *Respondent 12:* “This questionnaire would be well served by asking for an equation to be solved, then asking the exact steps taken to solve it as you simply remove five questions and aee given a full mental landscape runthrough in one text box. I suppose especially regarding the utilisation of features like “Perlin noise” it’s worth asking firstly what the game designeer or person believes perlin noise to be and how they came to that knowledge (for example, as an artist I use chat GPT advanced to transplate english programatical requirwmwnts to C++ with around a 90% efficiency rate which is crucial knowledge for a dissertation)”